Phase 4: LLM-as-a-Judge Evaluation

Experimental Design

To evaluate the generated SDBs, automated LLM judges are deployed along two distinct evaluation tracks:

- **Adapted RAG Triad Performance Audit:** This technical evaluation assesses the underlying technical retrieval performance of the three architectures that are tested in the experimental design. If uses an adapted RAG Triad rubric to score Faithfulness (Groundedness), Contextual Depth (Recall), and Interpretive Specificity (Answer Relevance)
- **SDB Adequacy Criteria:** This evaluation measures the scholarly and methodological completeness of the generated SDB, using the four adequacy criteria proposed in Section.

***Technical Evaluation Using the RAG Triad*.** Technical evaluation examines the performance of the three architectures using an adapted implementation of the RAG Triad. Because the objective of this study was corpus-level synthesis rather than record-level question answering, the conventional RAG Triad framework was modified to evaluate aggregate contextual representations rather than individual factual claims. The adapted evaluation considered three dimensions of technical performance (Fig. 1). Faithfulness assesses whether corpus-level characterizations presented in the generated sociocultural data biography are supported by the reference baseline established through the KBS audit. Context recall evaluates the extent to which the generated biography incorporates the structural characteristics of the primary corpus and, where applicable, the contextual knowledge contained in Notebook 2. Answer relevance evaluates the degree to which the generated biography synthesizes evidence into a coherent corpus-level interpretation rather than presenting disconnected observations or generic historical background. To adapt to the split notebook knowledge repositories (primary and context), the Context Recall metric in the Triad was decomposed into Primary Corpus and Sociocultural Context. This allowed evaluation of the extent to which the generated SDB incorporated sociocultural contextual knowledge, regardless of the mechanism by which that knowledge was acquired. Similarly, the Relevance metric was decomposed into Interpretive Synthesis and

Corpus Relevance. That permitted evaluation of the corpus-level synthesis, in addition to the grounding of outputs in the specific context of the primary corpus.

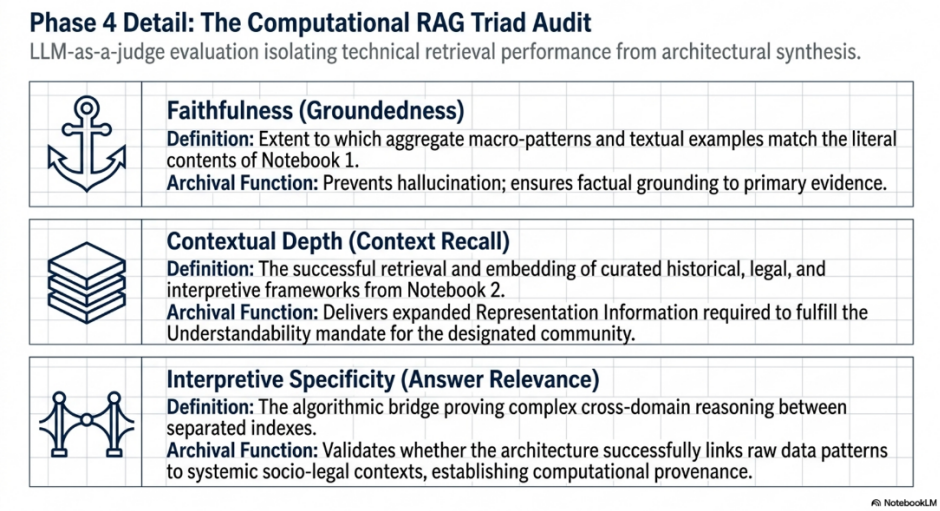


Fig. 1: Phase 4 Detail: The Computation RAG Triad Audit

Complete standardized prompts for the adapted RAG Triad evaluation, with incorporated scoring rubrics, are available in Appendix 3.

SDB Adequacy Evaluation. The second evaluation protocol used in the experiment is based on the operationalized adequacy criteria proposed in Phase 3. Whereas the adapted RAG Triad evaluates retrieval performance, the adequacy evaluation examines whether the generated outputs satisfy the methodological characteristics expected of an adequate sociocultural data biography. The adequacy evaluation therefore assesses the generated sociocultural data biography as a methodological artifact rather than as a retrieval system output. This distinction reflects the central objective of the study, which is to evaluate the operational production of sociocultural data biographies rather than retrieval performance alone.

Operationalization of the SDB Adequacy Criteria. The four SDB adequacy criteria are operationalized at three distinct levels in order to transition from an abstract universal theoretical framework into an evaluation instrument. Table 1 summarizes the three-level abstraction used to operationalize the adequacy criteria

Table 1: Three-level abstraction for operationalizing the SDB adequacy criteria

Level	Question Answered	Scope
Conceptual adequacy criteria	*What does this criterion mean?*	General to all sociocultural data biographies
Operational criteria	*What should an evaluator look for to determine whether the criterion has been satisfied?*	General evaluation guidance
Corpus-specific indicators	*What evidence in this corpus demonstrates that the operational criterion has been satisfied?*	Specific to the Domestic Trafficking Advertisements corpus

Level 1/Conceptual is the most abstract and portable definition of the SDB adequacy criteria. This level answers the question: ***What does the criterion mean?*** It describes what it means for an SDB to be contextually, evidentially, translationally, or analytically adequate in principle, independent of any particular dataset. These conceptual framings of the adequacy criteria can be considered universal benchmarks for any SDB, independent of the specific dataset or empirical domain being studied.

Level 2/Operational translates the conceptual criteria into explicit structural requirements that are study-specific. The operational criteria answer the question: **What would demonstrate that this criterion has been addressed in this experiment?** At this level, each conceptual criterion is translated into the specific aspects that an evaluator should look for.

- Operationally in this comparative experiment, **Contextual Adequacy** is instantiated through six analytical dimensions ***derived from systematic historical and archival analysis of the corpus and its contextual sources***: (1) record-generating purpose and institutional mandate; (2) socio-legal structures governing original information generation; (3) actors, perspectives, and authority; (4) representational conventions; (5) spatial and temporal variation in recording practices; and (6) structured absences,

evidentiary silences, and uncertainties — distinguished by type (accidental, practical, structural) with explicit identification of the mechanism producing each structural absence. Assessment further considers relational documentation: whether the SDB explains at least one data feature as the product of intersecting conditions across two or more dimensions, since some corpus features cannot be explained through any single dimension alone.

- **Evidentiary Adequacy** is operationalized via a decomposed scoring structure (separating primary corpus traceability, secondary context traceability, and reasoning transparency) to isolate reasoning quality from architectural index access. This is assessed along three dimensions: traceability (whether claims are grounded in specific, identifiable attributes in Notebook 1 or specific, named sources in Notebook 2, rather than asserted generically); transparency of reasoning (whether the inferential link between evidence and claim is made explicit); and uncertainty acknowledgment (whether fields subject to advertiser estimation or strategic misrepresentation, e.g., Age Specified, Skills Specified, are explicitly qualified as reflecting advertiser claims rather than verified fact).
- Operationalization of **Translational Adequacy** requires implementable guidance across four fixed computational decision categories (data preparation, concept operationalization, analytical design, and method selection), enforcing an ambiguity clause category discipline.
- Operationalization of **Analytical Utility** establishes a lifecycle requirement that explicitly checks for text redundancy and enforces a strict fabrication guardrail via a hypothetical framing integrity check.

Level 3/Corpus Specific is where the operational criteria get anchored to the actual schema and content. The question here is: **What evidence in this particular corpus would satisfy the operational criteria?** The criteria are fully instantiated for the specific empirical domain of this study: the Maryland State Archives Legacy of Slavery Domestic Trafficking Advertisements corpus**.** These indicators represent the explicit, observable data features and historical realities against which the generated outputs are audited.

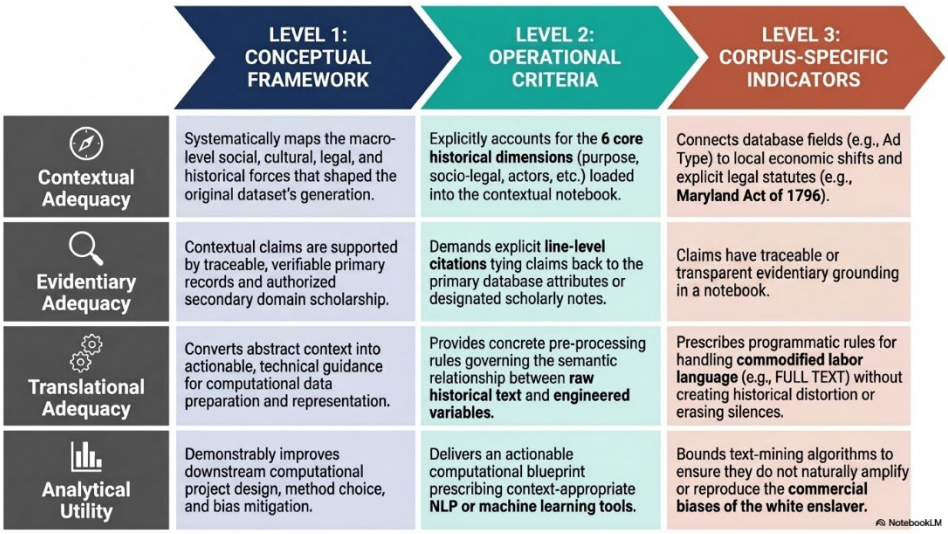


Fig. 2: Example of Operationalized Adequacy Criteria

Thus operationalization translates from general principles to indicators that instantiate those principles into a structured evaluation instrument specifically for the DTA corpus, as illustrated in Fig.2. Appendix 1(of the study) documents the comprehensive three-levels of abstraction for operationalizing the adequacy criteria for this comparative experiment using the DTA corpus. The comprehensive three-levels were developed iteratively, using three LLM assistants (Claude, ChatGPT, and Gemini).

LLM-as Judge. Rather than evaluating repeatability within a single judge model, this study evaluates the consistency of the proposed evaluation framework across three independent LLM judges: Gemini, ChatGPT, and Claude. Comparing outputs from different models is an approach used in recent LLM literature, to identify discrepancies or improve robustness. Each generated sociocultural data biography was evaluated independently by three different LLM-as-judge models using the same evaluation prompt and rubric. The purpose of using multiple judge models was to examine whether the proposed evaluation framework produced broadly similar evaluations across different language models. Scores and qualitative justifications were reported separately for each judge. No consensus procedure or composite score was calculated because differences among the judges were considered informative rather than treated as measurement error.

Manual Verification. Automated evaluation was supplemented by manual verification conducted by the researcher. This procedure was designed to identify classes of error that were outside the scope of the automated evaluation instruments, particularly errors involving attribution of evidence to specific records or contextual sources. as specified in the RAG Triad evaluation instrument.

Manual verification included spot-checking selected quotations, field values, summary statistics, and source citations against the primary corpus and the contextual knowledge base. Row-level citation accuracy in the Scenario 3 output received particular attention after review of that formal run identified a row-adjacency pattern: of three row-level citations spot-checked, two contained field values that were genuinely present in the corpus but misattributed to an adjacent record rather than the record actually cited. This is a retrieval- or synthesis-level error rather than a prompt-design flaw, and it does not necessarily affect corpus-level interpretive claims, but it does bear directly on the reliability of specific illustrative examples drawn from row-level citations. Such errors do not necessarily affect corpus-level interpretation but may reduce the reliability of specific illustrative examples.

Manual verification was conducted as a quality-control procedure rather than as an independent evaluation framework. Because verification was performed through targeted spot-checking rather than exhaustive review of every generated statement, it cannot establish the complete factual accuracy of the generated outputs. Instead, it provides additional evidence regarding the reliability of illustrative examples and source attribution beyond that obtainable through automated evaluation alone.

Comparative Analysis. Following completion of the technical evaluation, adequacy evaluation, and manual verification procedures, the three experimental conditions were compared qualitatively across both evaluation frameworks. Individual criterion scores, written justifications, and manual verification observations were considered together to identify similarities and differences among the generated sociocultural data biographies.

The purpose of the comparative analysis was not to rank the experimental conditions using a single numerical measure but to examine how differences in retrieval architecture influenced both technical retrieval performance and the methodological adequacy of the resulting

sociocultural data biographies. Differences among the independent judge models were retained and reported rather than averaged, providing additional evidence regarding the consistency with which the proposed evaluation framework could be applied across multiple language models.

4.5.8 Validity

This comparative experiment was designed to evaluate whether different retrieval architectures influence the generation of SDBs and whether those differences can be evaluated systematically using the SDB framework define in this study. The experimental design incorporates several procedures intended to strengthen the validity of these comparisons while acknowledging the limitations inherent in an exploratory study employing contemporary generative AI systems.

Internal validity. Internal validity concerns the extent to which observed differences among the generated SDBs can reasonably be attributed to experimental manipulation rather than to uncontrolled variation. The experiment was designed to compare three information-access configurations representative of practical retrieval environments. Because the configurations differ in more than one implementation characteristic, the study supports only comparison of the configurations as implemented, with particular attention to the incremental contribution of the contextual knowledge base between Conditions 2 and 3..

Nevertheless, the study implements additional controls to reduce potential sources of variation. For example, each experimental condition was conducted in a newly initiated LLM session to minimize contextual carryover between runs. The same fixed corpus and contextual knowledge base were used throughout formal data collection, and all generated outputs were evaluated using standardized evaluation procedures independent of the generation process.

Although these controls strengthen the internal validity of the study, they cannot eliminate variability associated with nondeterministic behavior in large language models or changes to cloud-based AI platforms over time. The findings should therefore be interpreted as observations of system behavior within the documented experimental environment rather than as immutable properties of the underlying models.

Construct validity. Construct validity concerns whether the experimental procedures adequately evaluate the methodological construct under investigation. In this study, the construct of interest

is the adequacy of the sociocultural data biography as a methodological artifact for computational treatment.

To support construct validity, the four adequacy criteria developed in Section 4.4 were translated into explicit operational criteria and corpus-specific indicators before formal experimentation began. These operational criteria informed both the design of the generation prompt and the evaluation framework, ensuring that all experimental conditions were evaluated against the same predefined methodological expectations.

The study does not claim that the operational criteria constitute validated measurement scales for the adequacy criteria. Rather, they provide a structured and transparent framework for evaluating the extent to which generated sociocultural data biographies exhibit the methodological characteristics proposed in this dissertation.

The findings of this study should be interpreted within the scope of the experimental corpus and software environment. The experiment employed a single historical corpus derived from the Maryland State Archives *Legacy of Slavery* collections together with a researcher-curated contextual knowledge base developed specifically for that corpus.

Consequently, the study does not establish that the proposed framework or the observed retrieval behavior will generalize to other archival collections, documentary genres, or retrieval-augmented generation platforms. Different historical domains may require different corpus-specific operational indicators, contextual knowledge bases, or retrieval implementations. Even so, the conceptual framework and the general approach to operationalization are intended to be transferable beyond the present case study. Future research should examine their applicability across additional archival collections, historical domains, and retrieval environments.

Interpretive validity. Interpretive validity concerns whether the conclusions drawn from the generated sociocultural data biographies are appropriately supported by the available evidence. Several features of the experimental design were intended to strengthen interpretive validity.

The Knowledge Base Sufficiency (KBS) pre-test audit established an independent reference baseline before any sociocultural data biographies were generated. Separate LLM-as-judge models were employed for technical and methodological evaluation, reducing the likelihood that

evaluation reflected the particular reasoning patterns of the generation model. In addition, manual verification of selected quotations, field values, and source attributions provided supplementary quality control for aspects of the generated outputs not addressed by the automated evaluation procedures.

Interpretive validity remains subject to the limitations of LLM-assisted evaluation. Although the evaluation prompts employed explicit operational criteria and structured scoring guidance, judgment necessarily involves interpretation. The written justifications accompanying each evaluation are therefore reported alongside the numerical scores to provide transparency regarding the reasoning supporting each assessment.